AI Evals for Engineers & PMs

Course syllabus · Instructor: Ion Moșnoi

A hands-on course on evaluating and improving AI products. Build a working agent, capture its behavior in traces, find its failures with a domain expert, and use evaluations to improve accuracy, cost and reliability.

Four weeks, with four live Saturday lectures, 10:00-12:00 Europe/Chișinău (UTC+2), plus 10+ hours of optional live office hours. Teaching is in English. Each module combines a lecture with a practical lab.

Module 1 - Building agents (Week 1)

Create the agent and traces that the rest of the course will use.

L1: AI fundamentals and the agent loop. Understand tokens, context limits, model behavior and structured outputs. Build a loop with tool calls, stop conditions and bounded retries.

Scope and permissions. Define allowed actions, when to ask for clarification and when to escalate to a human. Enforce permissions in code.

Make behavior visible. Record messages, model calls, tool inputs and outputs. Learn why a plausible final answer can hide a failed action.

Lab: Build a working agent and inspect its first complete trace.

Module 2 - Error analysis & evaluations (Week 2)

Turn expert reviews into an error taxonomy, synthetic test cases and calibrated LLM judges.

L2: Review and categorize failures. Inspect traces with a domain expert. Label Good or Bad, cite the failing step and expected behavior, then cluster notes by error type. Check frequency and impact; averages can hide a broken feature.

Create evaluation data. Combine features, roles and scenarios, including ambiguity and missing information. Review expected outcomes and evidence. Version datasets, check category coverage, and separate train, dev and held-out test cases.

One evaluator per failure mode. Define pass/fail rules. Use code for objective checks and LLM judges for interpretation, with clear criteria and diverse examples.

Calibrate against the expert. Inspect judge disagreements, false positives and false negatives. Improve the rubric on dev cases; report precision and recall by error type on untouched test cases.

Lab: Review traces, create a small synthetic dataset and calibrate one judge. Extend the taxonomy and evaluators in independent practice.

Module 3 - RAG that works (Week 3)

Build retrieval-augmented generation, expose its failures and measure each fix.

L3: Start with a naive RAG pipeline. Load documents, chunk, embed, store in a vector database, retrieve and generate. Run ten questions and collect weak answers as the starting point for improvements.

Chunking, embeddings and representation. Compare fixed, recursive, semantic and structure-aware chunking. Tune size and overlap; choose embeddings and useful metadata. Diagnose tables and scanned pages. Lab: re-chunk three ways and compare.

Measure retrieval separately. Create a 20-question golden set with relevant sources. Measure hit rate, recall@k and mean reciprocal rank (MRR). Score hybrid keyword/vector search, reranking and metadata filters against the same baseline; keep evidence-backed improvements.

Ground answers in evidence. Require citations, abstain when sources lack the answer and handle conflicting documents. Assess faithfulness, correctness and completeness separately from retrieval. Connect RAG to the agent as a tool; test citations and refusal.

Lab: Live: baseline, a guided comparison and grounded answers. Independent practice: three chunking variants, a 20-question set and three scored retrieval interventions.

Module 4 - Improve your agent (Week 4)

Use the failures to choose a fix, then measure what changed.

L4: Diagnose and experiment. Decide whether the problem is in data, retrieval, prompts, the model or tool flow. Try clearer instructions, better examples, query rewriting, routing or smaller task scopes.

Measure fixes and regressions. Compare versions on the same cases. Track failures by category alongside accuracy, cost and latency. Keep changes supported by the evidence.

Keep checking in production. Test prompt injection, permissions and unsafe actions. Automate regression checks in CI and monitor new failures and drift.

Lab: Run a before/after experiment and explain what improved, what regressed and what to try next.

Participation and support

Who it is for. Engineers and PMs building AI products. Basic Python is useful for engineers; PMs can pair for coding labs. Use the supplied project or authorized, anonymized company data.

Practice and support. Plan 3-5 hours a week for lectures and practice, plus optional office hours arranged with the cohort. Includes six assignments with solutions and walkthroughs, a 200+ page reader, lifetime recordings and materials, Discord access and future live cohorts. Complete four assignments, including the final report, for the completion certificate.

Cohorts and fees. 21 November-18 December 2026: €500. 19 December 2026-15 January 2027: €1,000. All prices are in EUR. Teams of 2-20 save 10%.

Your instructor. Ion Moșnoi has 11+ years of experience in AI/ML, helping teams from startups to enterprises improve AI products. He also delivers personalized two-day workshops for engineering teams.